

Name: Diya Ravindra Bhansali

PRN: 125B1F271

Class: SY Btech IT

Assignment 7: Implementing Exception Handling in Java Using In-Built Exceptions

CODE 1:

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        try
        {
            Scanner sc = new Scanner(System.in);
            System.out.println("Enter the number of elements :");
            int n = sc.nextInt();
            int[] arr = new int[n];
            System.out.println("Enter the elements:");
            for(int i = 0; i < n; i++)
            {
                arr[i] = sc.nextInt();
            }
            System.out.println("Enter the index of number you want to view:");
            int k = sc.nextInt();

            System.out.println("Value = " + arr[k]);

            System.out.println("Enter the element you want to divide the first element by:");
            int d = sc.nextInt();

            int ans = arr[0] / d;
            System.out.println("Answer = " + ans);

            String str = null;
            System.out.println(str.length());
        }
        catch(InputMismatchException e)
        {
            System.out.println("Value entered is not of required datatype");
        }
        catch(ArrayIndexOutOfBoundsException e)
        {
            System.out.println("Index you entered does not exist in Array");
        }
        catch(NegativeArraySizeException e)
        {
        }
```

```

        System.out.println("Array size cannot be negative");
    }
    catch(ArithmeticException e)
    {
        System.out.println("Number cannot be divided by 0");
    }
    catch(NullPointerException e)
    {
        System.out.println("Cannot access a null object");
    }
    catch(Exception e)
    {
        System.out.println("Some other exception occurred");
    }
    finally
    {
        System.out.println("Finally block executed");
    }
}
}

```

OUTPUT:

```

C:\Users\DIYA\Desktop>java Main
Enter the number of elements :
str
Value entered is not of required datatype
Finally block executed

```

```

C:\Users\DIYA\Desktop>java Main
Enter the number of elements :
5
Enter the elements:
1 2 3 4 5
Enter the index of number you want to view:
7
Index you entered does not exist in Array
Finally block executed

```

```

C:\Users\DIYA\Desktop>java Main
Enter the number of elements :
5
Enter the elements:
1 2 3 4 5
Enter the index of number you want to view:
8
Index you entered does not exist in Array
Finally block executed

```

```

C:\Users\DIYA\Desktop>java Main
Enter the number of elements :
5
Enter the elements:
1 2 3 4 5
Enter the index of number you want to view:
3
Value = 4
Enter the element you want to divide the first element by:
0
Number cannot be divided by 0
Finally block executed

```

```
C:\Users\DIYA\Desktop>java Main
Enter the number of elements :
5
Enter the elements:
1 2 3 4 5
Enter the index of number you want to view:
3
Value = 4
Enter the element you want to divide the first element by:
2
Answer = 0
Cannot access a null object
Finally block executed
C:\Users\DIYA\Desktop>_
```

CODE 2:

```
import java.util.*;
class Bank {
    int accno;
    String name;
    float bal;
    Bank(int a, String n, float b) {
        this.accno = a;
        this.name = n;
        this.bal = b;
    }
    void deposit(Scanner sc) throws Exception {
        System.out.println("Enter the amount you want to Deposit: ");
        float amt = sc.nextFloat();
        if (amt <= 0) {
            throw new Exception("Deposit amount must be greater than 0!");
        }
        bal = bal + amt;
        System.out.println("Total Balance = " + bal);
    }
    void withdraw(Scanner sc) throws Exception {
        System.out.println("Enter the amount you want to Withdraw: ");
        float amt = sc.nextFloat();
        if (amt <= 0) {
            throw new Exception("Withdrawal amount must be greater than 0!");
        }
        if (amt > bal) {
            throw new Exception("Insufficient Balance!");
        }
    }
}
```

$$\left. \begin{array}{l} \{ \\ \} \end{array} \right\}$$
[illegible]

```

        System.out.println("Account Added Successfully. ");
        System.out.println("Account Number = " + b[i].accno );
        i++;
        break;
    case 2:
        System.out.println("Enter Account Number to Remove: ");
        int r = sc.nextInt();
        if (r <= 0 || r > i || b[r - 1] == null) {
            throw new Exception("Account not present!");
        }
        b[r - 1] = null;
        System.out.println("Account Removed Successfully. ");
        break;
    case 3:
        System.out.println("Enter Account Number: ");
        int d = sc.nextInt();
        if (d <= 0 || d > i || b[d - 1] == null) {
            throw new Exception("Account not present!");
        }
        b[d - 1].deposit(sc);
        break;
    case 4:
        System.out.println("Enter Account Number: ");
        int w = sc.nextInt();
        if (w <= 0 || w > i || b[w - 1] == null) {
            throw new Exception("Account not present!");
        }
        b[w - 1].withdraw(sc);
        break;
    case 5:
        System.out.println("Enter the Account no:");
        int x = sc.nextInt();
        if (x <= 0 || x > i || b[x - 1] == null) {
            throw new Exception("Account not present!");
        }
        System.out.println("\nAccount no. " + b[x - 1].accno);
        System.out.println("Holder name: " + b[x - 1].name);
        System.out.println("Total Balance: " + b[x - 1].bal);
        break;
    case 6:
        System.out.println("\n**** Records ****");
        for (int k = 0; k < i; k++) {
            if (b[k] != null) {
                System.out.println("\nAccount no. " + b[k].accno);
                System.out.println("Holder name: " + b[k].name);
                System.out.println("Total Balance: " + b[k].bal);
            }
        }

```

```

        }
        break;
    case 7:
        System.out.println("Successfully Exited");
        break;
    default:
        throw new Exception("Invalid Choice!");
    }
} catch (Exception e) {
    System.out.println("Exception caught: " + e.getMessage());
}
} while (ch != 7);
} catch (Exception e) {
    System.out.println("Exception caught: " + e.getMessage());
} finally {
    System.out.println("Finally block executed. Thank you for using the banking system.");
}
}
}

```

OUTPUT:

```

C:\Users\DIYA\Desktop>java Main
Enter Account holder name: Ronak
Enter Initial Balance: 5000

**** MENU ****
1. Add Account
2. Remove Account
3. Deposit Money
4. Withdraw Money
5. View a specific Record
6. View all Records
7. Exit
Enter your choice: 3
Enter Account Number:
1
Enter the amount you want to Deposit:
2000
Total Balance = 7000.0

**** MENU ****
1. Add Account
2. Remove Account
3. Deposit Money
4. Withdraw Money
5. View a specific Record
6. View all Records
7. Exit
Enter your choice: 4
Enter Account Number:
1
Enter the amount you want to Withdraw:
8000
Exception caught: Insufficient Balance!

```

**** MENU ****

1. Add Account
2. Remove Account
3. Deposit Money
4. Withdraw Money
5. View a specific Record
6. View all Records
7. Exit

Enter your choice: 3

Enter Account Number:

5

Exception caught: Account not present!

**** MENU ****

1. Add Account
2. Remove Account
3. Deposit Money
4. Withdraw Money
5. View a specific Record
6. View all Records
7. Exit

Enter your choice: 7

Successfully Exited

Finally block executed. Thank you for using the banking system.

C:\Users\DIYA\Desktop>